

TRM-UNET: AN EFFICIENT EVENT-GUIDED MOTION DEBLURRING NETWORK

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ABSTRACT

Motion deblurring is a highly ill-posed problem, where event cameras provide complementary temporal cues that help resolve the inherent ambiguity. Although recent event-guided approaches have made remarkable progress, most methods focus on feature fusion in the encoder stage and rely on CNN or Transformer backbones, which either lack global context modeling or demand high computational costs. We present TRM-UNet, a compact yet powerful U-shaped architecture for event-based deblurring. Our design introduces Rotate Mamba blocks in the encoder to enable efficient long-range reasoning with linear complexity, together with a multilevel decoder that performs cross-modal fusion across multiple resolutions. Extensive experiments show that TRM-UNet achieves new state-of-the-art performance on the GoPro dataset while using only 64.1% of the parameters and 78.3% of the FLOPs compared to leading methods. The code and pre-trained model are available at: <https://github.com/NikonD850/TRM-UNet>.

Index Terms— Image restoration, motion deblurring, event camera, state space model, weighted fusion

1. INTRODUCTION

Motion deblurring, which aims to recover a sharp image from a blurry capture, remains a critically ill-posed problem in computer vision. The inherent ambiguity in inverting the motion blur process, where multiple latent sharp frames are convolved into a single measurement, has driven researchers to seek additional auxiliary data to constrain the solution space. Among various modalities, event cameras have emerged as a particularly powerful source of complementary information. Inspired by the peripheral structure of the primate retina, these sensors adopt a differential imaging model to asynchronously capture per-pixel brightness changes (known as ‘events’). This design abstracts the functional principles of the retinal periphery, enabling high temporal resolution and wide dynamic range in visual information acquisition. This unique capability provides a rich, continuous temporal prior of the motion that caused the blur, making event data a potent

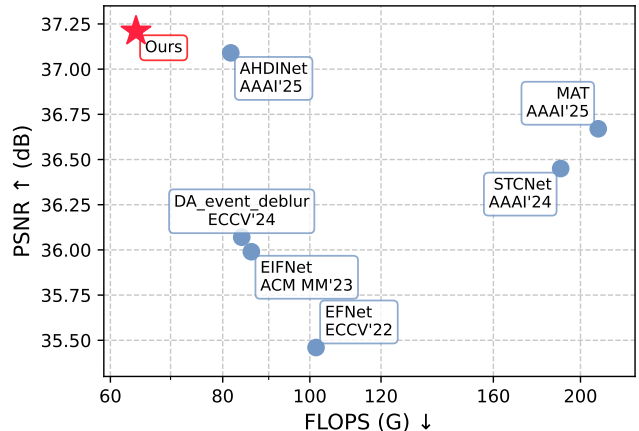


Fig. 1. Comparison with previous event-guided motion deblurring methods in terms of FLOPs-PSNR trade-off. FLOPs are calculated on a patch size of 256 with 6 event channels.

guide for the deblurring task. Consequently, several recent works have successfully leveraged events to achieve significant performance gains, establishing event-guided motion deblurring as a promising research direction.

We show several representative event-based deblurring networks in Fig. 1. EFNet [1] adopts two serial UNets [2], where the event features encoded by the first UNet are injected into the image encoder, and its high-level features further assist the second UNet. EIFNet [3] improves this design with a dual-encoder structure, cross-modal fusion, and a shared decoder. MAT [4] also employs a dual-encoder and single-decoder pipeline, where event features guide the image encoding stage in a manner similar to EFNet. More recently, AHDINet [5] introduced a dual-encoder and dual-decoder framework with final-stage fusion, currently achieving the highest Peak Signal-to-Noise Ratio (PSNR) among event-guided deblurring methods.

Although effective, these methods still leave space for improvement. First, most previous works primarily integrate event and image features in the encoder stage or at intermediate encoder-decoder levels, while largely overlooking the potential of cross-modal interaction in the decoder stage. Second, existing architectures are typically built on classical CNN [6] or Transformer [7] backbones, where CNNs struggle to capture long-range dependencies and Transformers suffer

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This work is supported by the CAS Project for Young Scientists in Basic Research (Grant No. YSBR-113).

from high computational costs, which prevents existing methods from maintaining high performance under limited computational budgets.

To address the first issue, we design a new twin-decoders architecture that performs cross-modal fusion at multiple resolutions, ensuring that the reconstruction process is consistently guided by event features. To address the second issue, we adopt State Space Models (SSMs), in particular the recently proposed Mamba [8], whose selective state mechanisms enable efficient global context modeling with linear computational complexity.

In this paper, we propose **TRM-UNet**, a lightweight yet powerful architecture that integrates Rotate Mamba blocks into a Twin UNet design for event-guided motion deblurring. Our main contributions are summarized as follows:

- We reconstruct the encoder using Rotate Mamba blocks to achieve efficient global context modeling.
- We design a multilevel decoder where cross-modal fusion is performed at multiple resolutions to effectively leverage event features.
- We achieve state-of-the-art performance on the Go-Pro dataset with significantly reduced parameters and FLOPs compared to prior methods.

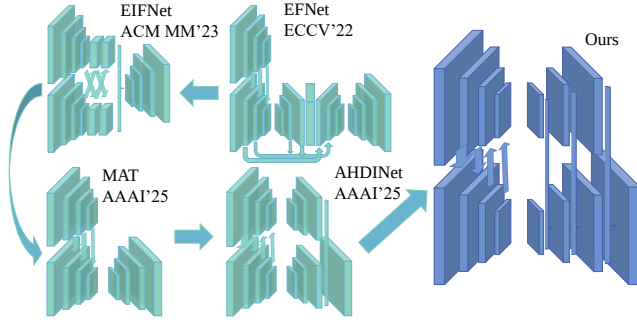


Fig. 2. Revisiting event-based deblurring networks.

2. METHOD

2.1. Overall Architecture

Following [5], we adopt a twin UNet architecture that integrates Event-assisted Edge Complement (EEC) and Image-assisted Semantic Complement (ISC) modules within the encoder to facilitate cross-modal feature enhancement at different levels of abstraction. Building upon this backbone, we further propose a novel multilayer fusion decoder, which progressively integrates event and image representations across multiple resolutions. This design not only strengthens the representational capacity of the network but also maintains high computational efficiency, making the overall framework lightweight and scalable. The encoder and decoder processes are formally expressed in Equation (1) and Equation (2),

where I and E denote the image and event streams, respectively; ‘EB’ and ‘DB’ refer to Encoder Block and Decoder Block; ‘WF’ represents Weighted Fusion; and $\downarrow$ denotes spatial downsampling.

$$\begin{aligned} I_1 &= \text{EB}_1^I(I_{\text{in}}), \quad E_1 = \text{EB}_1^E(E_{\text{in}}) \\ I_2 &= \text{EB}_2^I(\text{EEC1}(I_1^\downarrow, E_1^\downarrow)), \quad E_2 = \text{EB}_2^E(E_1^\downarrow) \\ I_3 &= \text{EB}_3^I(\text{EEC2}(I_2^\downarrow, E_2^\downarrow)), \quad E_3 = \text{EB}_3^E(E_2^\downarrow) \\ I_4 &= \text{EB}_4^I(I_3^\downarrow), \quad E_4 = \text{ISC}_4(I_4, \text{EB}_4^E(\text{ISC3}(I_3^\downarrow, E_3^\downarrow))) \end{aligned} \quad (1)$$

$$\begin{aligned} \hat{I}_3 &= \text{DB}_3^I(I_4, I_3), \quad \hat{E}_3 = \text{DB}_3^E(E_4, E_3) \\ \hat{I}_2 &= \text{DB}_2^I(\text{WF}_3(\hat{I}_3, \hat{E}_3), I_2), \quad \hat{E}_2 = \text{DB}_2^E(\hat{E}_3, E_2) \\ \hat{I}_1 &= \text{DB}_1^I(\text{WF}_2(\hat{I}_2, \hat{E}_2), I_1), \quad \hat{E}_1 = \text{DB}_1^E(\hat{E}_2, E_1) \\ I_{\text{out}} &= I_{\text{in}} + \text{Conv}(\text{WF}_1(\hat{I}_1, \hat{E}_1)) \end{aligned} \quad (2)$$

2.2. Encoder Design

The encoder blocks are constructed upon the Rotate Mamba (RM) module, which is derived from a modified version of the Efficient Visual State Space (EVSS) block [9]. To pursue a compact design, we significantly reduce the number of blocks per encoder layer, employing only 1 or 2 EVSS blocks instead of 6 to 12 blocks used in the original architecture. This reduction not only lowers the parameter count but also alleviates computational overhead while preserving the ability to model long-range dependencies. In order to exploit the shallow-layer skip connections of UNet and simultaneously capture global context at deeper levels, we introduce a 180° rotation operation at the input of each EVSS block. With this design, the output orientation remains consistent with the input after every two consecutive blocks, thereby preserving spatial correspondence for skip connections and ensuring stable information flow across encoder-decoder pathways. In particular, the first two encoder stages (EB_1 and EB_2) are each composed of two EVSS blocks, as formulated in Equation (3). This enables the network to learn multi-directional feature representations while maintaining strict spatial alignment with the input, a property that is crucial for effective feature fusion in subsequent decoding stages.

$$\text{Out} = \text{EVSS}_2(\text{EVSS}_1(\text{Conv}(\text{In}))) \quad (3)$$

In contrast, the deeper encoder layers (EB_3 and EB_4) utilize only one block, stated in Equation (4), facilitating the extraction of diagonally-oriented and global features during skip connections.

$$\text{Out} = \text{EVSS}(\text{Conv}(\text{In})) \quad (4)$$

2.3. Decoder Design

As illustrated in Equation (2), we introduce three weighted fusion blocks to effectively integrate high-level representa-



Fig. 3. Visual results and relative PSNR measures on the GoPro [10] test set.

tions from both image and event branches. This cross-modal feature guidance mechanism enables multiscale features from the event branch to actively steer the image feature upsampling process. Specifically, the event features $\hat{E}_3$ and $\hat{E}_2$ at different scales are fused with the corresponding image features $\hat{I}_3$ and $\hat{I}_2$ through weight-based adaptive fusion. This allows high-frequency details from event data to enhance the reconstruction of image features while maintaining semantic consistency. The fused representations are then processed by the image decoder blocks DB_2^I and DB_1^I along with skip connections from the encoder (I_2 and I_1), enabling precise spatial recovery and detail restoration. The structure of the weighted fusion block is formulated as:

$$\begin{aligned}
 A &= \hat{I}_{in} + \hat{E}_{in} \\
 P &= AP(A) + MP(A) \\
 W_I, W_E &= \text{split}(\sigma(\text{Conv}(\text{ReLU}(\text{Conv}(P)))))) \\
 \text{Out} &= W_I \odot \hat{I}_{in} + W_E \odot \hat{E}_{in}
 \end{aligned} \quad (5)$$

where AP and MP denote average pooling and max pooling, respectively, and $\odot$ represents the Hadamard product. The decoder blocks retain the same structure as in [5].

2.4. Loss Function

We employ the Charbonnier loss [11] for end-to-end training, consistent with the original configuration of [5]. The loss is defined as:

$$\mathcal{L}_{\text{char}} = \frac{1}{C \times H \times W} \sum \sqrt{(\hat{I} - I_{GT})^2 + \epsilon^2} \quad (6)$$

where $\hat{I}$ denotes the predicted image, I_{GT} is the ground truth image, $C \times H \times W$ represents the volume of the image tensor, and ϵ is set to 10^{-3} .

3. EXPERIMENTS

3.1. Dataset and Pre-processing

To ensure a fair comparison, all methods are trained and validated on the widely used GoPro dataset [10]. The dataset comprises 3214 pairs of blurry and corresponding sharp images with a resolution of 1280×720 pixels, divided into 2103 training pairs and 1111 testing pairs. It was constructed using a high-frame-rate GoPro Hero Black camera recording at 240 fps. Sharp images are extracted as the central frames of short video sequences, while blurry images are synthesized by averaging consecutive frames to simulate motion blur. In accordance with [3, 12, 5], events are simulated using ESIM [13] with 6 channels.

3.2. Experimental Settings

Our method is implemented in PyTorch 2.5.1 and trained on NVIDIA GeForce RTX 3090 GPUs with CUDA 12.4. The models are trained for 2×10^5 iterations using a patch size of 512 and a batch size of 4, following the configuration of [5]. The Adam optimizer is employed with a cosine annealing learning rate schedule, starting from 2×10^{-4} and decaying to 10^{-6} . Data augmentation includes random cropping and random horizontal flipping with a probability of 0.5. For testing, previous methods are evaluated using official implementations and pre-trained models or provided result images.

3.3. Results

To demonstrate the superiority of the proposed approach, we compare TRM-UNet with several state-of-the-art deblurring methods: EFNet [1], REFID [14], EIFNet [3], STCNet [12], DANet [15], MAT [4], AHDINet [5], ConvIR [16], EA-Mamba [17], and MaIR [18]. It is noteworthy that during the reproduction of previous works, we observed discrepancies in the reported Structure Similarity Index Measure (SSIM) [19] values. Specifically, when evaluating using the official SSIM implementation on either the Y channel or all three channels, the results did not align with those reported in the original publications. For instance, our method achieves an SSIM value of **0.9709** on the Y channel, whereas AHDINet [5] yields 0.9706 compared to its reported 0.9820, and STCNet [12] achieves 0.9619 versus a reported 0.9809. To ensure a fair comparison, we employ PSNR as the primary evaluation metric in this section.

Visual Results. Fig. 3 presents a visual comparison with recent image restoration methods (highlighted in blue) and event-based motion deblurring approaches. In the upper example, our model successfully restores the character shapes on the left and accurately reconstructs the color of the woman’s hair on the right. In the lower example, our method is the only one capable of reconstructing the boundary gap between the trunk and the roof, demonstrating the effectiveness of our approach.

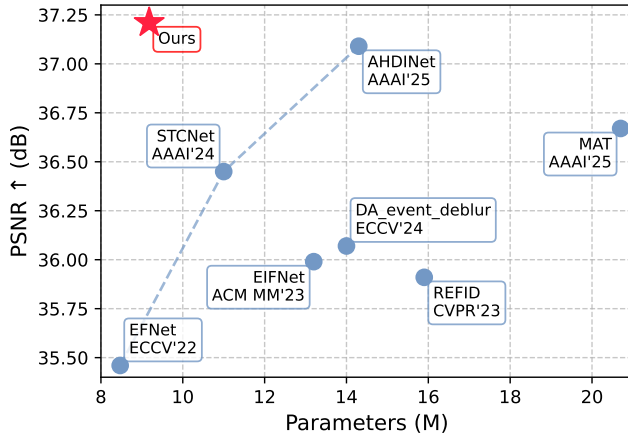


Fig. 4. Comparison with previous event-guided motion deblurring methods in terms of the parameters-PSNR trade-off.

Efficiency. Our model achieves superior performance in both FLOPs-PSNR and parameters-PSNR comparisons. As shown in the FLOPs-PSNR plot (Fig. 1), our model attains a PSNR improvement of 0.12 dB over previous models (**37.21** dB vs 37.09 dB), while reducing computational complexity by 21.7% (**64.07** GFLOPs vs. 81.67 GFLOPs). This indicates a significant enhancement in efficiency without sacrificing performance. In the parameters-PSNR plot (Fig. 4), our method occupies the upper-left region, signifying that it achieves the highest PSNR with the fewest parameters among

compared methods. This underscores the model’s ability to balance parameter efficiency and reconstruction quality effectively, making it suitable for practical applications.

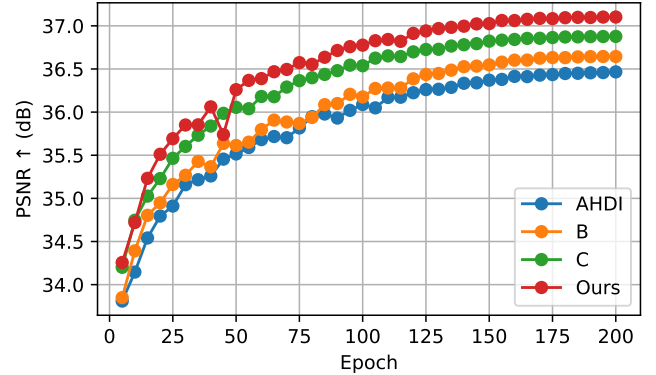


Fig. 5. PSNR progression during training across different model configurations.

3.4. Ablation Study

To evaluate the effectiveness of our contributions in a lightweight setting, we performed ablation studies using a patch size of 384. In Table 1, ‘Enc’ denotes the encoder, ‘Dec’ the decoder, ‘AHDINet’ refers to AHDINet [5], ‘Ori’ indicates the original method, ‘RMO’ represents Rotate MambaOut [20], ‘RM’ denotes Rotate Mamba, and ‘MF’ stands for Multi-Fusion. The comparison between AHDINet and configuration B illustrates the improvements brought by the new encoder architecture. The difference between B and C highlights the impact of the Mamba module, while the transition from C to our full model demonstrates the advantages of the proposed multi-fusion decoder. Fig. 5 shows the increasing trend of PSNR during training, further validating the effectiveness of our improvements.

Table 1. Ablation study of component (PS=384).

Net	Enc	Dec	Para(M)↓	FLOP(G)↓	PSNR↑
AHDINet	Ori	Ori	14.32	81.67	36.47
B	RMO	Ori	9.086	64.69	36.65
C	RM	Ori	9.162	64.07	36.88
Ours	RM	MF	9.180	64.07	37.10

4. CONCLUSION

In this paper, we introduce TRM-UNet, a novel event-based deblurring network with a reconstructed encoder implemented Mamba and a new designed decoder implemented image-event feature fusion in multiple layers. Our method achieves a +0.12 PSNR improvement while using only 64.1% of the parameters and 78.3% of the FLOPs compared to the runner-up method, setting a new SOTA on the widely used GoPro dataset [10].

5. REFERENCES

- [1] Lei Sun, Christos Sakaridis, Jingyun Liang, Qi Jiang, Kailun Yang, Peng Sun, Yaozu Ye, Kaiwei Wang, and Luc Van Gool, “Event-based fusion for motion deblurring with cross-modal attention,” in *European conference on computer vision*. Springer, 2022, pp. 412–428.
- [2] Olaf Ronneberger, Philipp Fischer, and Thomas Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *International conference on medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.
- [3] Wen Yang, Jinjian Wu, Leida Li, Weisheng Dong, and Guangming Shi, “Event-based motion deblurring with modality-aware decomposition and recomposition,” in *Proceedings of the 31st ACM international conference on multimedia*, 2023, pp. 8327–8335.
- [4] Senyan Xu, Zhijing Sun, Mingchen Zhong, Chengzhi Cao, Yidi Liu, Xueyang Fu, and Yan Chen, “Motion-adaptive transformer for event-based image deblurring,” in *Proceedings of the AAAI conference on artificial intelligence*, 2025, vol. 39, pp. 8942–8950.
- [5] Wen Yang, Jinjian Wu, Leida Li, Weisheng Dong, and Guangming Shi, “Asymmetric hierarchical difference-aware interaction network for event-guided motion deblurring,” in *Proceedings of the AAAI conference on artificial intelligence*, 2025, vol. 39, pp. 9265–9273.
- [6] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 2002.
- [7] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin, “Attention is all you need,” *Advances in neural information processing systems*, vol. 30, 2017.
- [8] Albert Gu and Tri Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.
- [9] Lingshun Kong, Jiangxin Dong, Jinhui Tang, Ming-Hsuan Yang, and Jinshan Pan, “Efficient visual state space model for image deblurring,” in *Proceedings of the computer vision and pattern recognition conference*, 2025, pp. 12710–12719.
- [10] Seungjun Nah, Tae Hyun Kim, and Kyoung Mu Lee, “Deep multi-scale convolutional neural network for dynamic scene deblurring,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 3883–3891.
- [11] Pierre Charbonnier, Laure Blanc-Feraud, Gilles Aubert, and Michel Barlaud, “Two deterministic half-quadratic regularization algorithms for computed imaging,” in *Proceedings of 1st international conference on image processing*. IEEE, 1994, vol. 2, pp. 168–172.
- [12] Wen Yang, Jinjian Wu, Jupao Ma, Leida Li, and Guangming Shi, “Motion deblurring via spatial-temporal collaboration of frames and events,” in *Proceedings of the AAAI conference on artificial intelligence*, 2024, vol. 38, pp. 6531–6539.
- [13] Henri Rebecq, Daniel Gehrig, and Davide Scaramuzza, “Esim: an open event camera simulator,” in *Conference on robot learning*. PMLR, 2018, pp. 969–982.
- [14] Lei Sun, Christos Sakaridis, Jingyun Liang, Peng Sun, Jiezhong Cao, Kai Zhang, Qi Jiang, Kaiwei Wang, and Luc Van Gool, “Event-based frame interpolation with ad-hoc deblurring,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 18043–18052.
- [15] Zhijing Sun, Xueyang Fu, Longzhuo Huang, Aiping Liu, and Zheng-Jun Zha, “Motion aware event representation-driven image deblurring,” in *European conference on computer vision*. Springer, 2024, pp. 418–435.
- [16] Yuning Cui, Wenqi Ren, Xiaochun Cao, and Alois Knoll, “Revitalizing convolutional network for image restoration,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 46, no. 12, pp. 9423–9438, 2024.
- [17] Yu-Cheng Lin, Yu-Syuan Xu, Hao-Wei Chen, Hsien-Kai Kuo, and Chun-Yi Lee, “Eamamba: Efficient all-around vision state space model for image restoration,” *arXiv preprint arXiv:2506.22246*, 2025.
- [18] Boyun Li, Haiyu Zhao, Wenxin Wang, Peng Hu, Yuanbiao Gou, and Xi Peng, “Mair: A locality-and continuity-preserving mamba for image restoration,” in *Proceedings of the computer vision and pattern recognition conference*, 2025, pp. 7491–7501.
- [19] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli, “Image quality assessment: from error visibility to structural similarity,” *IEEE transactions on image processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [20] Weihao Yu and Xinchao Wang, “Mambaout: Do we really need mamba for vision?,” in *Proceedings of the computer vision and pattern recognition conference*, 2025, pp. 4484–4496.